



HKBK COLLEGE OF ENGINEERING

Nagawara, Bangalore-560045



Department of Computer Science & Engineering
Project Presentation
VI semester

BCV654A –Water conservation & Rain water Harvesting

Topic:Water alert domestic use and rainwater calculator

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Introduction & Problem Statement

Introduction

- Water resources are becoming increasingly scarce due to overuse and climate change.
- Identifying underground water sources is challenging using traditional methods.
- Lack of awareness and proper tools leads to inefficient water usage and poor conservation.
- Existing systems do not provide integrated analysis, visualization, and decision support

Problem Statement

How to effectively detect underground water resources and promote water conservation using modern digital tools and geophysical methods

Core Features

- ▶ Real-time Data Visualization
- ▶ Water Usage Monitoring System
- ▶ Alert System for Excess Water Consumption
- ▶ Integration of Geophysical Methods
- ▶ Analytical Dashboard
- ▶ User Login & Admin Panel

Benefits and Impact

- ▶ Promotes efficient water management
- ▶ Helps in identifying underground water sources
- ▶ Reduces water wastage through alerts
- ▶ Supports sustainable development
- ▶ Provides data-driven decision making
- ▶ Increases awareness about water conservation
- ▶ Useful for agriculture, urban planning, and rural areas

Geophysical Methods Overview

Geophysical methods are used to detect underground water without digging.

Based on physical properties like electrical resistivity, seismic waves, and magnetic fields.

Types:

- ▶ Electrical Resistivity Method
- ▶ Seismic Method
- ▶ Ground Penetrating Radar (GPR)Magnetic Method

Purpose:

- ▶ Locate groundwater
- ▶ Analyze soil and rock structure

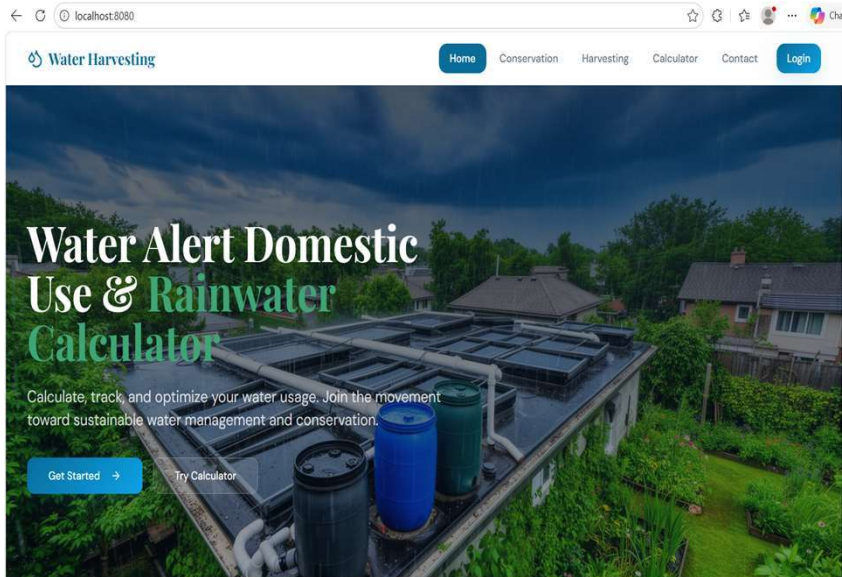
Project Overview

- ▶ This project is a Smart Water Management & Detection System.
- ▶ Combines Geophysical Analysis + Web Technology.
- ▶ Helps users monitor water usage and detect underground water.
- ▶ Provides a user-friendly interface for data interpretation.
- ▶ Includes analytics, alerts, and visualization tools.

Data Visualization & Analytics

- ▶ Pie charts for water usage distribution
- ▶ Line graphs for daily consumption trends
- ▶ Comparison charts for methods
- ▶ Location-based data insights
- ▶ Alert indicators for excessive usage
- ▶ Dashboard provides real-time monitoring

Result(Snap Shots)

A screenshot of a "Create Account" form. At the top, there is a green circular icon with a white water drop and a leaf. Below the icon, the heading "Create Account" is followed by the subtext "Join the rainwater harvesting community". The form is enclosed in a light gray border and contains the following sections: "Select Role" with two buttons labeled "User" and "Admin"; "Full Name" with a text input field containing the placeholder "Your full name"; "Email" with a text input field containing the placeholder "you@example.com"; "Password" with a text input field containing the placeholder "Create a password (min 6 chars)"; and "Confirm Password" with a text input field containing the placeholder "Confirm your password". At the bottom of the form is a large green "Create Account" button. Below the button, there is a link that says "Already have an account? Sign In".

- ▶ The system successfully monitors water usage
- ▶ Provides accurate visualization and analytics
- ▶ Helps in identifying potential groundwater locations
- ▶ Improves decision-making efficiency

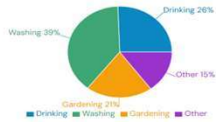
Dashboard

Track your water usage and rainfall data.

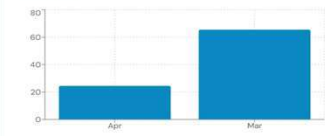
Log Rainfall Log Water Usage



Water Usage Breakdown



Monthly Rainfall



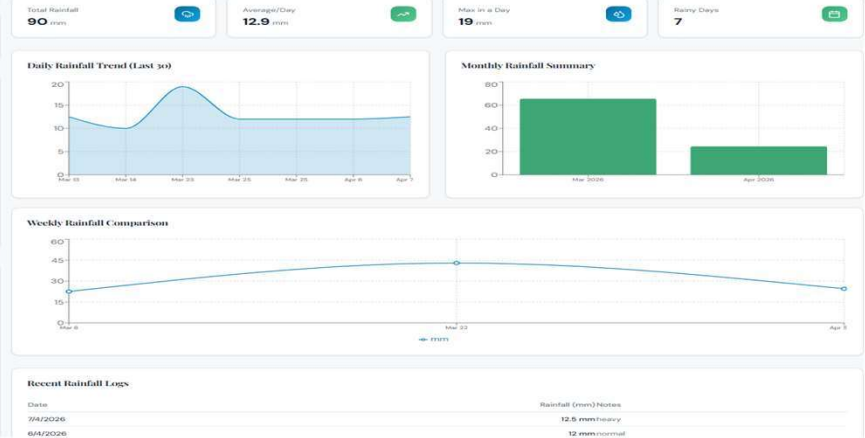
Recent Water Usage Entries

Date	Drinking	Washing	Gardening	Other	Total	Action
3/4/2026	20L	60L	12L	2L	94L	
3/4/2026	20L	60L	20L	40L	140L	
6/4/2026	14L	19L	2L	2L	52L	
6/4/2026	20L	50L	10L	5L	85L	
6/4/2026	23L	12L	12L	4L	51L	
25/3/2026	8L	8L	15L	4L	25L	

Rainfall Data Tracker

Monitor and analyze your local rainfall patterns.

Log Rainfall



Water Saving Tips

Practical tips to reduce your water consumption.

Daily Water Usage Limit

Limit: 135L - Edit

Today's usage: **234L / 135L**

- ⚠️ Daily Limit Exceeded!
- You've used 234L today, exceeding your 135L limit by 99L. Your highest category is **washing**.
- Take shorter showers - aim for under 5 minutes.
 - Only run the washing machine with full loads.
 - Turn off the tap while scrubbing dishes or brushing teeth.
 - Fix any leaking faucets immediately.

Tip of the Day - General

Monitor your water bill for unexpected increases that may signal hidden leaks.

AI-Powered Personalized Tips

Get My Tips

Click "Get My Tips" to receive personalized water-saving recommendations based on your usage data.



Water Alert Domestic Use & Rainwater Calculator

Calculate, track, and compare your water usage with the movement toward sustainable water management and conservation.

Get Started

Calculator

Why Water Harvesting?

Everything you need to manage water harvesting and conservation efficiently.

- Track Water Usage:** Log daily water consumption by category and visualize your usage patterns with pie charts.
- Smart Calculator:** Calculate potential domestic demand based on your roof area and local rainfall data.
- Visual Analytics:** View reports and graphs to monitor savings, rainfall trends, and environmental impact.
- Secure & Private:** Your data is protected with authentication and role-based access control.

Rainwater Calculator

Estimate how much rainwater you can harvest based on your roof area, surface type, and local rainfall.

Input Parameters

Roof / Catchment Area (m²)

0.85

Annual Rainfall (mm)

499

Surface Type

Tiled / RCC Rooftops (0.75 – 0.95)

Runoff Coefficient

0.85

Range for Tiled / RCC Rooftops: 0.75 – 0.95

Calculate

Reset

Result

361

liters per year

That's approximately 1 liters per day

Save to My Account

Formula

Volume (L) = Catchment Area (m²) × Rainfall (mm) × Runoff Coefficient

1 mm of rain on 1 m² = 1 liter of water. The runoff coefficient depends on the surface type.

Surface Types & Runoff Coefficients

Surface Type	Min Coefficient	Max Coefficient	Average
Tiled / RCC Rooftops	0.75	0.95	0.85
Concrete / Paved Surfaces	0.7	0.9	0.80
Rocky Surfaces	0.6	0.8	0.70
Bare Soil	0.1	0.4	0.25
Agricultural Fields	0.2	0.5	0.35
Grass-covered Land	0.1	0.3	0.20

Admin Panel

Overview

Manage Users

Rainfall Data

Rainfall Tracker

Collections

Water Usage

Water Tips & Insights

Back to App

Logout

Water Harvesting – Admin Overview

System-wide statistics and management.

Total Users

2

Rainfall Logs

10

Rainfall Tracker

View

Collections

0

Water Usage

12

Tips & Insights

View

Source Code

```
# index.css 5 • .gitignore bun.lockb components.json 1 index.html package.json 1 postcss.config.js
rain-harvest-manager-main > TS tailwind.config.ts > ...
1 import type { Config } from "tailwindcss";
2
3 export default {
4   darkMode: ["class"],
5   content: ["/pages/**/*.ts,tsx", "/components/**/*.ts,tsx", "/app/**/*.ts,tsx", "/src/**/*.ts,tsx"],
6   prefix: "",
7   theme: {
8     container: {
9       center: true,
10      padding: "2rem",
11      screens: {
12        "2xl": "1400px",
13      },
14    },
15    extend: {
16      fontFamily: {
17        display: ["Playfair Display", "serif"],
18        body: ["DM Sans", "sans-serif"],
19      },
20      colors: {
21        border: "hsl(var(--border))",
22        input: "hsl(var(--input))",
23        ring: "hsl(var(--ring))",
24        background: "hsl(var(--background))",
25        foreground: "hsl(var(--foreground))",
26        primary: {
27          DEFAULT: "hsl(var(--primary))",
28          foreground: "hsl(var(--primary-foreground))",
29        },
30        secondary: {
31          DEFAULT: "hsl(var(--secondary))",
32          foreground: "hsl(var(--secondary-foreground))",
33        },
34        destructive: {
35          DEFAULT: "hsl(var(--destructive))",
36          foreground: "hsl(var(--destructive-foreground))",
37        },
38        muted: {
39          DEFAULT: "hsl(var(--muted))",
40          foreground: "hsl(var(--muted-foreground))",
41        },
42        accent: {
43          DEFAULT: "hsl(var(--accent))",
44          foreground: "hsl(var(--accent-foreground))",
45        },
46        popover: {
47          DEFAULT: "hsl(var(--popover))",
48          foreground: "hsl(var(--popover-foreground))",
49        },
50        card: {
```

```
# index.css 5 • .gitignore bun.lockb components.json 1
rain-harvest-manager-main > package.json > ...
1 {
2   "name": "vite_react_shadcn_ts",
3   "private": true,
4   "version": "0.0.0",
5   "type": "module",
6   "scripts": {
7     "dev": "vite",
8     "build": "vite build",
9     "build:dev": "vite build --mode development",
10    "lint": "eslint .",
11    "preview": "vite preview",
12    "test": "vitest run",
13    "test:watch": "vitest"
14  },
15  "dependencies": {
16    "@hookform/resolvers": "^3.10.0",
17    "@radix-ui/react-accordion": "^1.2.11",
18    "@radix-ui/react-alert-dialog": "^1.1.14",
19    "@radix-ui/react-aspect-ratio": "^1.1.7",
20    "@radix-ui/react-avatar": "^1.1.10",
21    "@radix-ui/react-checkbox": "^1.3.2",
22    "@radix-ui/react-collapsible": "^1.1.11",
23    "@radix-ui/react-context-menu": "^2.2.15",
24    "@radix-ui/react-dialog": "^1.1.14",
25    "@radix-ui/react-dropdown-menu": "^2.1.15",
26    "@radix-ui/react-hover-card": "^1.1.14",
27    "@radix-ui/react-label": "^2.1.7",
28    "@radix-ui/react-menubar": "^1.1.15",
29    "@radix-ui/react-navigation-menu": "^1.2.13",
30    "@radix-ui/react-popover": "^1.1.1 Loading...",
31    "@radix-ui/react-progress": "^1.1.7",
32    "@radix-ui/react-radio-group": "^1.3.7",
33    "@radix-ui/react-scroll-area": "^1.2.9",
34    "@radix-ui/react-select": "^2.2.5",
35    "@radix-ui/react-separator": "^1.1.7",
36    "@radix-ui/react-slider": "^1.3.5",
37    "@radix-ui/react-slot": "^1.2.3",
38    "@radix-ui/react-switch": "^1.2.5",
39    "@radix-ui/react-tabs": "^1.1.12",
40    "@radix-ui/react-toast": "^1.2.14",
41    "@radix-ui/react-toggle": "^1.1.9",
42    "@radix-ui/react-toggle-group": "^1.1.10",
43    "@radix-ui/react-tooltip": "^1.2.7",
44    "@supabase/supabase-js": "^2.98.0",
45    "@tanstack/react-query": "^5.83.0",
46    "class-variance-authority": "^0.7.1",
47    "clsx": "^2.1.1",
48    "cmdk": "^1.1.1",
49    "date-fns": "^3.6.0",
50    "date-fns-jal": "^3.0.0",
51    "embla-carousel-react": "^8.3.0",
52    "framer-motion": "^11.11.17",
53    "lucide-react": "^0.454.0",
54    "react": "^18.3.1",
55    "react-dom": "^18.3.1",
56    "react-hook-form": "^7.53.1",
57    "react-use": "^17.5.0",
58    "recharts": "^2.15.0",
59    "sonner": "^1.7.0",
60    "tailwind-merge": "^2.5.4",
61    "tailwindcss": "^3.4.15",
62    "tailwindcss-animate": "^1.0.7",
63    "vite": "^5.4.11",
64    "vite-plugin-devtools": "^7.4.0",
65    "vite-plugin-eslint": "^1.1.0",
66    "vite-plugin-react": "^3.1.0",
67    "vite-plugin-svgr": "^4.2.0",
68    "vite-tsconfig-paths": "^4.3.2",
69    "vitest": "^2.1.8",
70    "zod": "^3.24.1"
71  },
72  "devDependencies": {
73    "@eslint/js": "^9.17.0",
74    "@types/react": "^18.3.12",
75    "@types/react-dom": "^18.3.1",
76    "@types/react-use": "^17.0.0",
77    "@types/testing-library__jest-dom": "^5.14.9",
78    "@types/uuid": "^10.0.0",
79    "autoprefixer": "^10.4.20",
80    "eslint": "^9.17.0",
81    "eslint-plugin-react": "^7.37.2",
82    "eslint-plugin-react-hooks": "^4.6.2",
83    "eslint-plugin-tailwindcss": "^3.18.0",
84    "eslint-plugin-unicorn": "^56.0.1",
85    "eslint-plugin-unused-imports": "^3.1.0",
86    "globals": "^15.14.0",
87    "jest": "^29.7.0",
88    "jest-environment-jsdom": "^29.7.0",
89    "jest-transform-stub": "^2.0.0",
90    "postcss": "^8.4.49",
91    "postcss-import": "^16.1.0",
92    "postcss-js": "^4.0.1",
93    "postcss-load-config": "^4.0.2",
94    "postcss-nested": "^6.0.1",
95    "postcss-selector-parser": "^6.1.2",
96    "prettier": "^3.4.2",
97    "prettier-plugin-tailwindcss": "^0.6.9",
98    "tailwindcss": "^3.4.15",
99    "typescript": "^5.7.2",
100   "typescript-eslint": "^8.19.0",
101   "vite": "^5.4.11",
102   "vite-plugin-devtools": "^7.4.0",
103   "vite-plugin-eslint": "^1.1.0",
104   "vite-plugin-react": "^3.1.0",
105   "vite-plugin-svgr": "^4.2.0",
106   "vite-tsconfig-paths": "^4.3.2",
107   "vitest": "^2.1.8",
108   "zod": "^3.24.1"
109 }
110
```

```

# index.css 5
rain-harvest-manager-main > src > # index.css > root
1  @import url('https://fonts.googleapis.com/css2?family=DM+Sans:ital,opsz,wght@0,9..40,100..1000;1,9..40,100..1000');
2
3  @tailwind base;
4  @tailwind components;
5  @tailwind utilities;
6
7  @layer base {
8    :root {
9      --background: 200 20% 97%;
10     --foreground: 210 50% 10%;
11
12     --card: 0 0% 100%;
13     --card-foreground: 210 50% 10%;
14
15     --popover: 0 0% 100%;
16     --popover-foreground: 210 50% 10%;
17
18     --primary: 199 89% 32%;
19     --primary-foreground: 0 0% 100%;
20
21     --secondary: 152 45% 45%;
22     --secondary-foreground: 0 0% 100%;
23     --muted: 200 20% 94%;
24     --muted-foreground: 210 15% 45%;
25
26     --daccet: 152 45% 92%;
27     --accent-foreground: 152 45% 20%;
28
29     --destructive: 0 72% 51%;
30     --destructive-foreground: 0 0% 100%;
31
32     --border: 200 20% 88%;
33     --input: 200 20% 88%;
34     --ring: 199 89% 32%;
35
36     --radius: 0.75rem;
37
38     --sidebar-background: 210 50% 8%;
39     --sidebar-foreground: 200 20% 90%;
40     --sidebar-primary: 199 89% 48%;
41     --sidebar-primary-foreground: 0 0% 100%;
42     --sidebar-accent: 210 40% 15%;
43     --sidebar-accent-foreground: 200 20% 90%;
44     --sidebar-border: 210 30% 18%;
45     --sidebar-ring: 199 89% 48%;
46
47     /* Custom tokens */
48     --water-gradient: linear-gradient(135deg, hsl(199, 89%, 32%), hsl(199, 89%, 48%));
49     --eco-gradient: linear-gradient(135deg, hsl(152, 45%, 45%), hsl(152, 55%, 55%));
50     --hero-overlay: linear-gradient(180deg, hsla(210, 50%, 8%, 0.7) 0%, hsla(210, 50%, 8%, 0.4) 100%);

```

```

# index.css 5  .gitignore bun.lockb {} components.json index.html
rain-harvest-manager-main > index.html > ...
1  <!doctype html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8" />
5    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6    <!-- TODO: Set the document title to the name of your application -->
7    <title>Lovable App</title>
8    <meta name="description" content="Lovable Generated Project" />
9    <meta name="author" content="Lovable" />
10
11    <!-- TODO: Update og:title to match your application name -->
12    <meta property="og:title" content="Lovable App" />
13    <meta property="og:description" content="Lovable Generated Project" />
14    <meta property="og:type" content="website" />
15    <meta property="og:image" content="https://lovable.dev/opengraph-image-p98pqg.png" />
16
17    <meta name="twitter:card" content="summary_large_image" />
18    <meta name="twitter:site" content="@Lovable" />
19    <meta name="twitter:image" content="https://lovable.dev/opengraph-image-p98pqg.png" />
20  </head>
21
22  <body>
23    <div id="root"></div>
24    <script type="module" src="/src/main.tsx"></script>
25  </body>
26 </html>
27

```

Conclusion

- ▶ The project provides an effective solution for water conservation and groundwater detection.
- ▶ Integration of geophysical methods enhances accuracy.
- ▶ The system improves awareness and efficient usage of water resources.
- ▶ Future improvements can include AI-based predictions and IoT integration.

THANK YOU

